A Local Scale–Filtered Interface for Atomic Short–Range Effects

A Consistency Analysis within the MMU Framework

J. Wollbold

Abstract

We demonstrate that atomic short–range effects such as hyperfine splitting, the Lamb shift, and effective screening cannot be consistently described by point–like interactions. Within the Methane Metauniverse (MMU) framework, these effects necessarily arise from a local overlap region (S–node) acting as a scale–filtered interface between two interacting cells. Using dimensional analysis, reduced–mass scaling, and transfer tests (H $\to \mu$H $\to \mu$D), we prove that the interface possesses an intrinsic length scale $\xi(\mu) \propto \mu^{-1}$. We show that contact interactions are ruled out by muonic hydrogen data, and that a geometrically filtered interface is mathematically unavoidable.

1 Introduction

Precision spectroscopy of hydrogenic systems reveals several well–known short–range effects: hyperfine splitting (21 cm line), the Lamb shift, and effective charge screening. In conventional QED these arise from distinct perturbative mechanisms. In the MMU framework, space itself is an elastic medium composed of discrete cells with internal degrees of freedom (w_2, w_3, w_4). Local interactions occur only in overlap regions between cells, termed *S–nodes*.

The purpose of this paper is to demonstrate that:

1. these effects share a single local origin,
2. point–like interactions are inconsistent with muonic systems,
3. the S–node must act as a scale–filtered interface,
4. its characteristic length ξ is geometrically fixed.

2 MMU Assumptions

We explicitly assume:

(A1) Physical interactions occur only through local overlap of MMU cells.

(A2) Each cell is governed by a fixed elastic K–matrix $(k_2, k_3, k_4, k_{23}, k_{24}, k_{34})$.

(A3) Observable quantities arise after elimination of internal degrees of freedom (Schur reduction / projection).

(A4) No new fundamental length scales may be introduced beyond those constructed from $(\hbar, c, \alpha, \mu)$.

3 Proposition I: Necessity of a Local Interaction Node

Proposition. Atomic short–range effects must originate from a localized interaction region.

Proof. Hyperfine splitting, Lamb shift, and screening all vanish when the electronic wavefunction does not overlap the nucleus. They therefore cannot arise from bulk cell dynamics. In MMU geometry, overlap is localized to S–nodes. □

4 Proposition II: Failure of Point–Like Interfaces

Proposition. A point–like S–node (contact interaction) is inconsistent with muonic hydrogen.

Proof. For a contact interaction,

$$\Delta E \propto |\psi(0)|^2 \propto \mu^3. \tag{1}$$

Between hydrogen and muonic hydrogen, μ increases by a factor ≈ 186, implying an enhancement $\sim 6.4 \times 10^6$. Experimentally, the muonic Lamb shift is only $\sim 4 \times 10^4$ larger. Thus contact interactions overpredict the shift by orders of magnitude. □

5 Proposition III: Existence of a Scale–Filtered Interface

Proposition. The S–node must suppress contributions from arbitrarily short scales.

Proof. Since point–like coupling is excluded (Proposition II), the interface must possess an intrinsic spatial extent ξ that filters high–momentum (short–wavelength) contributions. Such filtering must be physical, not a numerical cutoff. □

6 Proposition IV: Geometric Determination of $\xi(\mu)$

Proposition. If the interface is purely geometric, its length scale satisfies

$$\xi(\mu) = \eta \frac{\hbar}{\mu c \alpha}, \tag{2}$$

with η dimensionless.

Proof. By assumption (A4), ξ may depend only on $\hbar, c, \alpha, \mu$. Dimensional analysis (Buckingham–Π theorem) shows that the only combination with dimensions of length is $\hbar/(\mu c)$. Since α is dimensionless, it may appear only multiplicatively. No other dependence is permitted without introducing a new scale. □

7 Proposition V: Physical Meaning of Reduced Mass

Proposition. The reduced mass μ represents the inertia of the local S–node deformation.

Proof. In the two–body problem, transformation to center–of–mass and relative coordinates yields a kinetic term $\frac{1}{2}\mu\dot{r}^2$. In MMU geometry, r parametrizes deformation of the overlap region. Thus μ measures the resistance of the S–node to deformation. □

8 Proposition VI: Unified Origin of Short–Range Effects

Proposition. Hyperfine splitting, Lamb shift, and screening are different orders of the same local interface physics.

Proof. Elimination of internal MMU degrees of freedom yields:

- first–order effects in w_3 (spin torsion): hyperfine splitting,

- second–order effects in coupled w_3–w_4: Lamb shift,
- static renormalization of w_2: screening.

All arise from the same S–node K–matrix structure. □

9 Discussion

The analysis demonstrates that a local, scale–filtered interface is not an interpretive choice but a mathematical necessity. Muonic systems provide a decisive kill test excluding contact interactions. The MMU framework naturally accommodates this through geometric filtering, without renormalization counterterms.

10 Conclusion

We have proven that:

1. atomic short–range effects require a local interaction node,
2. point–like interfaces are ruled out by experiment,
3. the interface must be scale–filtered,
4. its scale is uniquely fixed by geometry,
5. reduced mass has a direct physical meaning,
6. multiple precision effects share a single origin.

The spacetime deformation discussed here is elastic and local within the MMU framework and must not be confused with general-relativistic curvature.

The present work establishes internal consistency for hydrogenic two-body systems; multi-nucleon geometries require additional structure and are left for future work.

This establishes a consistent, falsifiable geometric foundation for atomic short–range physics within the MMU framework.

Remark on the MMU Project

Author's Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model's conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every nontrivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI's standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

References

[1] J. Wollbold. *A Geometric–Fractal Tetrahedral Model of Matter and Space: From Atomic Bonds to Cosmic Structures.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/HYAX5.

[2] J. Wollbold. *From Geometry to Spectra: Predicting Atomic Masses and Ionization Energies from Space Elasticity in the MMU Framework.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/JC2S4.

[3] J. Wollbold. *Geometric Origin of the 21 cm Line and Hyperfine Spin-Flip Structure in the Methane Metauniverse.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/JKV6H.

[4] J. Wollbold. *Unified Determination of the Cross-Coupling Constants k_{23}, k_{24} and k_{34} in the MMU Tetrahedral Model from Larmor, Zeeman and Lamb Resonances.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/48EQX.

[5] J. Wollbold. *Determination of the UR–Tetrahedron in the Methane Metauniverse (MMU): From Elastic Constants to Lepton Scaling.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/NRW3S.

[6] J. Wollbold. *Dissipative Lagrange Field Equation of the Methane Metauniverse (MMU): Geometric Origin of Thermal Damping and Spatial Healing.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/VUTNX.

[7] J. Wollbold. *The Methane Metauniverse (MMU): A Case Study in AI-Assisted Theoretical Physics and Human–Machine Scientific Creativity.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/SCXPQ.

[8] J. Wollbold. *Geometric Quantization of the Internal Edge Length in the MMU Model.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/NUPXY.

[9] J. Wollbold. *MMU 2.0: A Unified Geometric Model of Internal Spin, Torsion–Volume Coupling, and the g–Factor in Dual Tetrahedral Spacetime.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/N754E.

[10] J. Wollbold. *Geometric Derivation of the Planck Scale in the Methane Metauniverse (MMU).* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/42K9S.

[11] J. Wollbold. *Discovery of an Electro–Gravity Fixed Point Beyond the Planck Scale in the MMU Tetrahedral Model*, OSF Preprint (2025), https://doi.org/10.17605/OSF.IO/ACQNH.

[12] J. Wollbold. *From Length–Dependent to Mass–Dependent Stiffness in the MMU: $k_i(a) \rightarrow k_i(m)$ and Leptonic Balmer Predictions.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/V43AY.

[13] J. Wollbold. *Projection Principle for Quantum Orbitals: Angular Momentum, Spin, and the Pauli Exclusion Rule as Elastic Torsion in the MMU Tetrahedral Model.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/863MV.

[14] J. Wollbold. *The Dirac Equation as the External Projection of the MMU Ur-Cell: A Geometric Origin of Spinors, Antiparticles, and Relativistic Dynamics.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/HG9FN.

[15] J. Wollbold. *Geometric Quantization of Space and the Origin of the Rydberg Constant in the MMU.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/RJXE5.

[16] J. Wollbold. *Electromagnetic Constants from Tetrahedral Geometry: A Geometric Derivation of ε_0, μ_0, Z_0 and c in the Methane Metauniverse (MMU).* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/AB4F2.

[17] J. Wollbold. *Lepton Masses, Koide Geometry, and the $g-2$ Family Pattern from Tetrahedral Torsion in the Methane Metauniverse (MMU).* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/GU9RQ.

[18] J. Wollbold. *Proton Chair Eigenmodes in the Methane Metauniverse (MMU): From Internal Rotor Frequencies to Hydrogenic Space Response.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/BKDCP.

[19] J. Wollbold. *Gravitationally Induced Shifts in Atomic Spectra: A Testable Prediction of Tetrahedral Spacetime Geometry.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/U8B4E.

[20] J. Wollbold. *Microscopic Spacetime Kinematics and the Emergence of Einstein Gravity from the Tetrahedral Geometry of the MMU.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/W96H3.

[21] J. Wollbold. *MMU 3.0: A Minimal Geometric–Elastic Spacetime Model and the Emergence of Quantum, Relativistic, and Electromagnetic Structure.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/BDTA6.

[22] J. Wollbold. *Why Matter Requires Antimatter: A Stability Mechanism in the Dual–Tetrahedral MMU Geometry.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/JYRHK.

[23] J. Wollbold. *A Unified Geometric Prediction of Balmer, Zeeman and Lamb Isotope Shifts.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/MZRQ3.

[24] J. Wollbold. *Neutron–Proton Mass Gap as an Electro–Gravity Phase Transition in the MMU.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/SD58Q.

[25] J. Wollbold. *A Geometric Proton–Quark Model and Proton g–Factor from Chair–Quark Torsion in the Methane Metauniverse.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/F5NGS.

[26] J. Wollbold. *Lepton Masses from Torsional Geometry of Space and Their Verification via Hyperfine and Isotope Shifts.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/P8ZT7.

[27] J. Wollbold. *Black Hole Horizons and Hawking Temperature as Projection Phenomena in the Methane Metauniverse.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/M5TH3.

[28] J. Wollbold. *Geodesics and Gravitational Redshift as Projection Necessities in the Methane Metauniverse.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/JKXY7.

[29] J. Wollbold. *Unavoidability of the Dirac Operator from Dual–Tetrahedral Geometry in the Methane Metauniverse.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/Y3MQ5.

[30] J. Wollbold. *Wavefunction, Linearity, and Measurement as Lattice Dynamics in the Methane Metauniverse.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/WG48D.

[31] J. Wollbold. *A Minimal Geometric–Elastic Origin of Quantum Dynamics, Lorentz Symmetry and Weak Decay Rates in the Methane Metauniverse (MMU).* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/TREHB.

[32] J. Wollbold. *Elastic Prestress, Time, and Gravitation in the Methane Metauniverse Framework.* OSF Preprint (2025). DOI: https://doi.org/10.17605/OSF.IO/2QRMS.